Grade 6 Accelerated
Unit 14
Lesson 10 Homework

Name_Answer Key_________
Date____________________
Period___________________

1. When flying on an airplane, your travel bag can weigh no more than 50 lbs. Your bag currently weighs 27.3 lbs. The inequality $27.3 + m \leq 50$ represents the situation, where m represents the amount of weight that can be added to your bag. What is the maximum amount of weight that can be added to your bag while still making the weight requirement?

$$\begin{array}{rcl} 27.3 + \mathrm{m} & \leq & 50 \\ \underline{-27.3 \quad\quad} & & \underline{-\ 27.3} \\ m & \leq & 22.7 \end{array}$$

No more than 22.7 lbs. can be added to your bag.

2. You earn $\$8.75$ per hour for walking dogs in your neighborhood. You are saving up your dog walking money to purchase a new kayak that costs $\$280$. The inequality $\$8.75h \geq \$280$ represents the situation, where h represents the number of hours worked. What is the minimum number of hours you must walk in order to purchase the kayak?

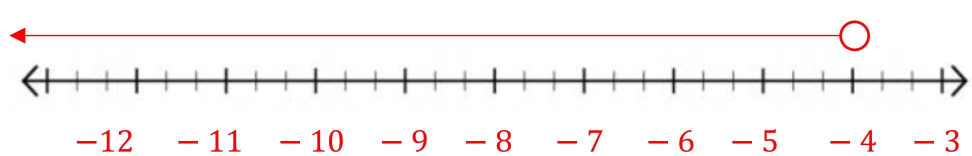
$$\frac{8.75\mathrm{h}}{8.75} \geq \frac{280}{8.75}$$
$$h \geq 32$$

You must work at least 32 hours in order to purchase the kayak.

3. Complete the table by substituting each value of x to determine if that is a solution to each inequality.

Value of x	**What is the value of $-4.5x$**	**Is $-4.5x > 18$ true or false?**	**Solution?**
-4	18	$18 > 18$	o Yes o No
-1.5	6.75	$6.75 > 18$	o Yes o No
3	-13.5	$-13.5 > 18$	o Yes o No
$6\frac{1}{5}$	-27.9	$-27.9 > 18$	o Yes o No

4. Graph the inequality on the number line.



5. Find the solution to both inequalities using the division property of inequality.

$\mathbf{-7y \leq -35}$ $\frac{-7y}{-7} \leq \frac{-35}{-7}$ $y \geq 5$	$\mathbf{7y \leq -35}$ $\frac{7y}{7} \leq \frac{-35}{7}$ $y \leq -5$

6. Find the solution to both inequalities using the multiplication property of inequality.

$\mathbf{\frac{y}{3} > 2}$ $3 \cdot \frac{y}{3} > 2 \cdot 3$ $y > 6$	$\mathbf{\frac{y}{-3} > 2}$ $-3 \cdot \frac{y}{-3} > 2 \cdot -3$ $y < -6$

Solve each inequality. Then, graph its solution set.

7. $\frac{r}{-4} < 7$

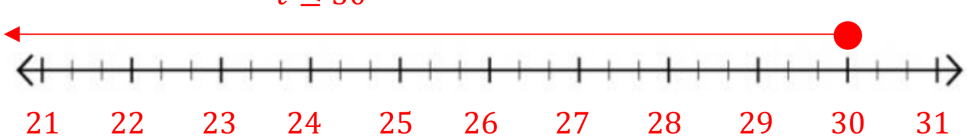
$-4 \cdot \frac{r}{-4} < 7 \cdot -4$

$r > -28$

8. $-0.4t \geq -12$

$\frac{-0.4t}{-0.4} \geq \frac{-12}{-0.4}$

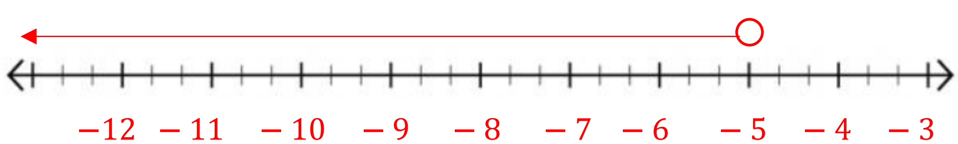
$t \leq 30$



9. $\frac{m}{-2} > \frac{5}{2}$

$-2 \cdot \frac{m}{-2} > \frac{5}{2} \cdot -2$

$m < -5$



10. Explain why the direction of the inequality symbol must be reversed when multiplying or dividing by the same negative number.

Because when you multiply or divide by a negative number, the value becomes its opposite (sign wise) and so the opposite sign is used.